

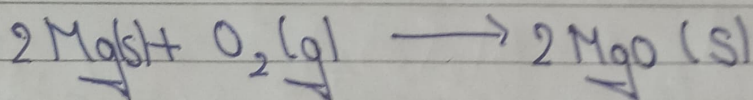
Chemical Reactions And Equations

Text questions

1. Activity 1.1

What $\rightarrow$ do you observe?

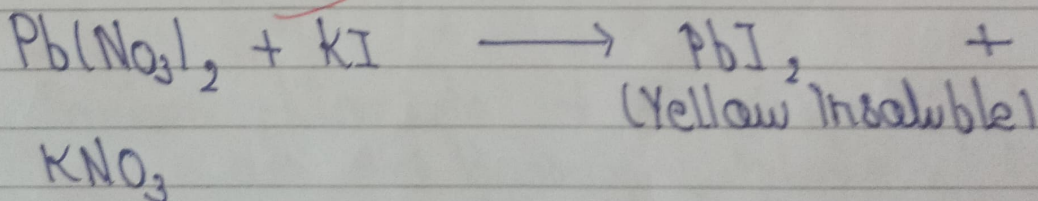
Sol: The magnesium ribbon reacts with oxygen present in air to form magnesium oxide



2. Activity 1.2

What $\rightarrow$ do you observe?

When lead nitrate and Potassium iodide solution is mixed they react vigorously and produce lot's of heat with an insoluble substance ~~formed~~.

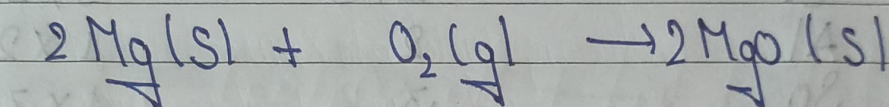


Questions

P.g: 6

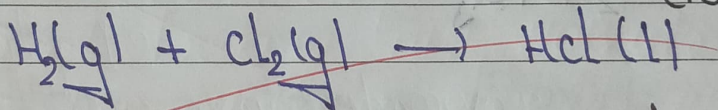
1. Why should a magnesium ribbon be cleaned before burning in air?

Solution: Magnesium ribbon reacts with oxygen present in air/atmosphere to form a layer of magnesium oxide and it affects combustion.



2. Write the chemical equation for the following reactions and balance them.

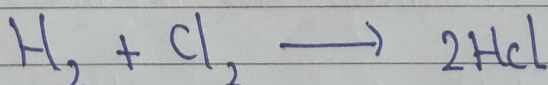
(i) Hydrogen + chlorine $\rightarrow$ Hydrogen chloride.



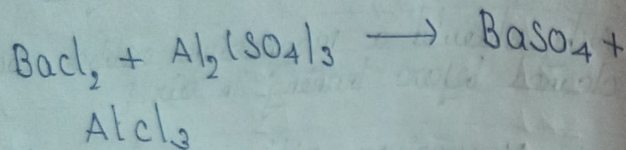
<u>E</u>	<u>Reactants</u>	<u>Products</u>
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H	2	1 x 2 = 2
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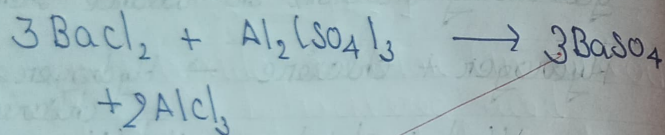
Cl	2	2 x 1 = 2
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(ii) Barium chloride + Aluminium sulphate $\rightarrow$ Barium sulphate + Aluminium chloride



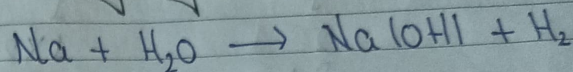
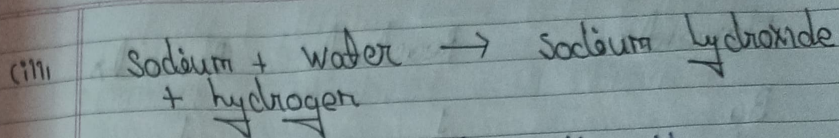
Elements	Reactants	Products
Ba	1x3	1x3
Cl	2x3	3x2
Al	2	1x2
So ₄	3	1x3



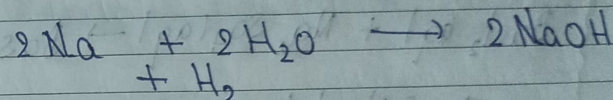
{ Hence, the no. of atoms on both the sides are equal then it is balanced... }

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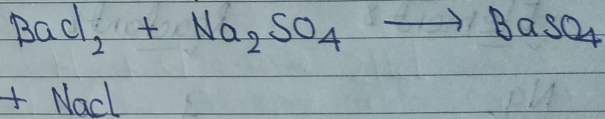


Elements	Reactants	Products
Na	1x2	1x2
H	2x2	1+2=3
O	1x2	1x2

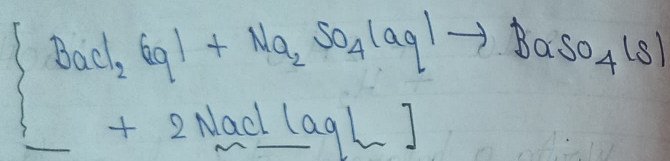
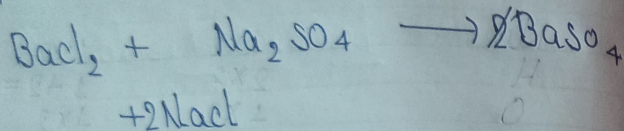


3. Write a balanced chemical equation with state symbols for the following reactions.

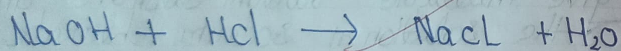
(i) Solutions of barium chloride and sodium sulphate in water react to give insoluble barium sulphate and the solution of sodium chloride



Elements	Reactants	Products
Ba	1	1
Cl	2	1 x 2
Na	2	1 x 2
SO ₄	1	1



(ii) Sodium hydroxide solution (in water) reacts with hydrochloric acid solution (in water) to produce sodium chloride solution and water.



Elements	Reactants	Products
Na	1	1
OH	1	1
H	2	2
Cl	1	1

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Hence, the no. of atoms on both the sides are equal it is already balanced

2. Activity 17

What happens in each case? / Which gas is present in each test tube?

Sol. Oxygen is present at the anode and hydrogen at cathode

Ratio of hydrogen to oxygen is 2:1

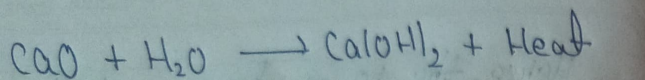
Questions

1. A solution of a substance 'X' is used for white washing

(i) Name the substance 'X' and write its formula

Substance which is used for white washing is calcium oxide, CaO

(ii) Write the reaction of the substance 'X' named in (i) above with water.



<u>Elements</u>	<u>Reactants</u>	<u>Products</u>
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Ca	1	1
O	2	2
H	2	2

It's already balanced

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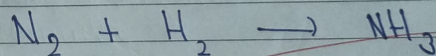
Exercises

4. What is a balanced chemical equation? why should chemical equations be balanced?

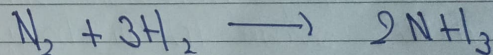
- Chemical equation in which the atoms of elements are equal on both sides.
- Chemical equation should be balanced because of law of conservation of mass it states that mass can neither be created nor be destroyed.

5. Translate the following statements into chemical equations and then balance them.

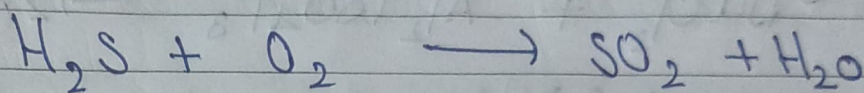
(a) Hydrogen gas combines with nitrogen to form ammonia.



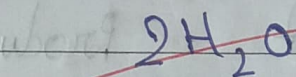
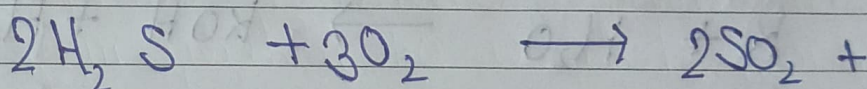
<u>Elem</u>	<u>Reactants</u>	<u>Products</u>
N	2	1
H	2 x 3	3 x 2



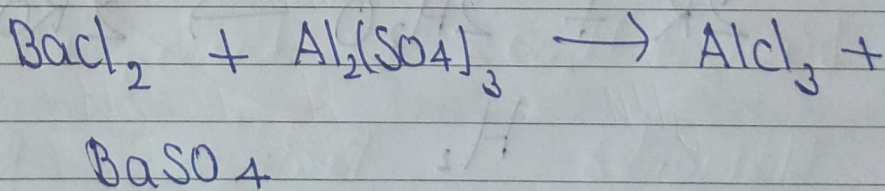
- (b) Hydrogen sulphide gas burns in air to give water and sulphur dioxide.



<u>Elem</u>	<u>React</u>	<u>Prods</u>
H	2 x 2	12 x 2
S	1 x 2	1 x 2
O	2 x 3	3 x 2



- (c) Barium chloride reacts with aluminium sulphate to give aluminium chloride + and a precipitate of barium sulphate.

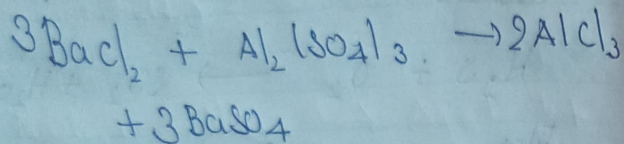


Elements	Reactants	Products
Ba	1	1
Cl	2	3
Al	2	1

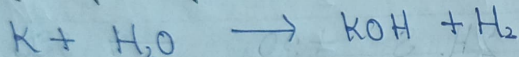
SO₄

3

1



(d) Potassium metal reacts with water to give potassium hydroxide and hydrogen gas



Elements	React	Products
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K

1

1

H

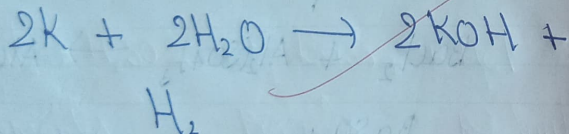
2

3

O

1

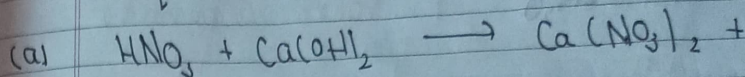
1



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6. Balance the following chemical equations.



H₂O

Elements

Reactants Products

H

1+2

2

N

1x2

2

O

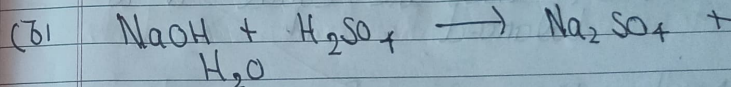
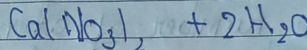
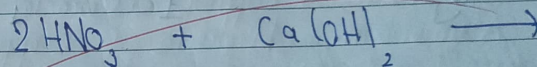
5

6+1

Ca

1

1



H₂O

Elements

Products

Reactants

Na

1

2

O

5

5

H

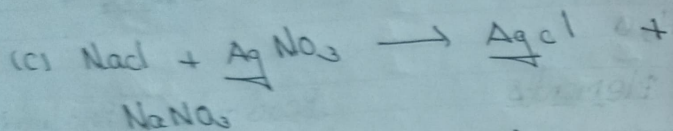
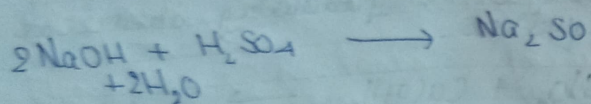
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2

SO₄

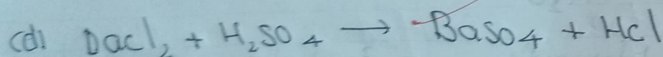
1

1

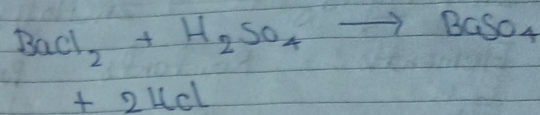


Elements	Reactants	Products
Na	1	1
Cl	1	1
Ag	1	1
NO ₃	1	1

Hence, the number of atoms on both sides are equal & is balanced.

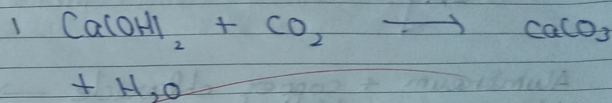


Elements	Products	Reactants
Ba	1	1
Cl	2	1
H	2	1
SO ₄	1	1



7. Write the balanced chemical equations following reactions

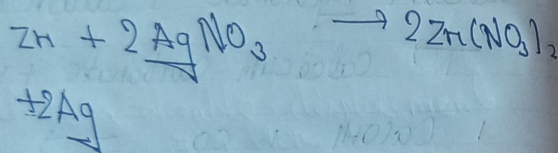
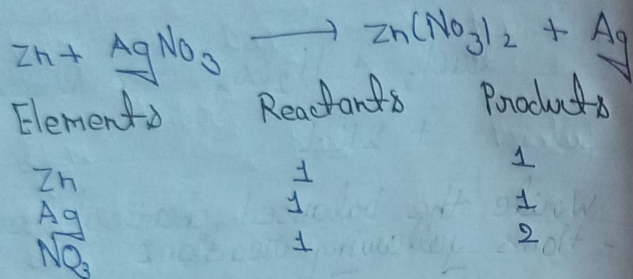
(a) Calcium hydroxide + carbon dioxide
 $\longrightarrow$ Calcium carbonate + water



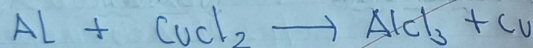
Elements	Products	Reactants
Ca	1	1
OH	2+2	4
C	1	1
H	2	2

Hence the number of atoms on both the sides are equal & is already balanced.

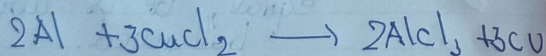
(b) Zinc + Silver nitrate $\longrightarrow$ Zinc nitrate + silver



(c) Aluminium + copper chloride → Aluminium chloride + copper



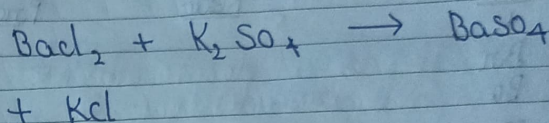
Elements	Reactants	Products
Al	1	1
Cu	1	1
Cl	2	3



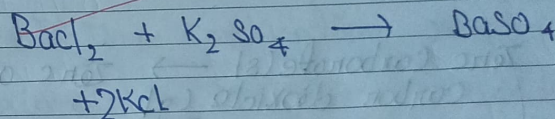
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(d) Barium chloride + Potassium sulphate
→ Barium sulphate + Potassium chloride

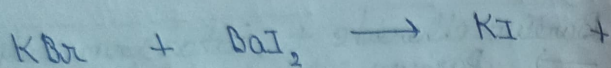


Elements	Reactants	Products
Ba	1	1
Cl	2	2
K	2	2
SO ₄	1	1



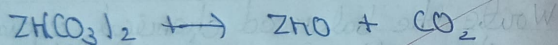
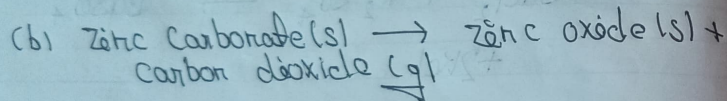
8. Write the balanced chemical equation for the following and identify the type of reaction in each case.

(a) Potassium bromide (aq) + Barium iodide (aq) → Potassium iodide (aq) + barium bromide (s)



Elem	React	Product
K	1	1
Br	1	1
Ba	1	1
I	2	2

It is a double-displacement reaction.

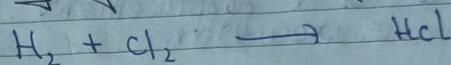
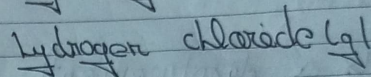
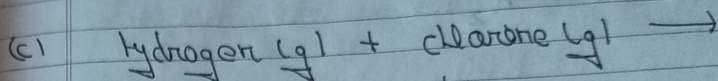


Elem	Reactants	Products
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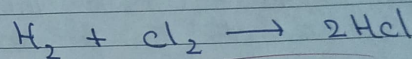
Zn	1	1
C	1	1
O	3	3

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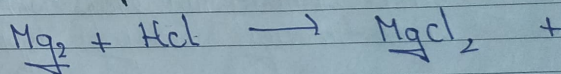
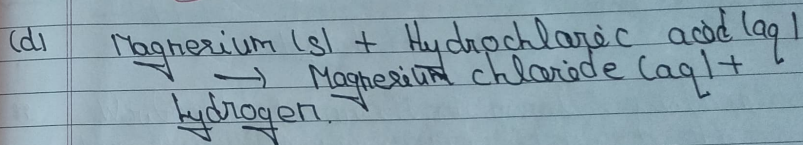
It is a decomposition reaction.



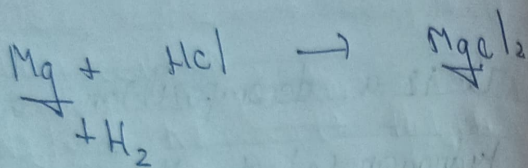
Elem	Rea	Product
H	2	2
Cl	2	2



It is a combination reaction.



Elements	Rea	Products
Mg	1	1
H	2	2
Cl	2	2



It is a displacement reaction.

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9. What does mean by exothermic and endothermic reactions?
Give examples.

Sol: Reactions in which heat is given out along with the products are called exothermic reactions.

Reactions in which energy is absorbed are known as endothermic reactions.

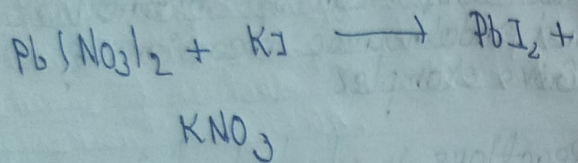
10. Why is respiration considered an exothermic reaction? Explain.

Sol: Respiration is considered as exothermic reaction as heat is given out during the process.

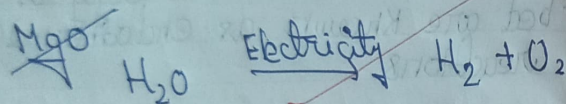
11. Why are decomposition reactions called the opposite of combination reactions? Write equations for these reactions.

As in decomposition reactions compound splits into two compounds or elements as in combination reaction two or more elements or compounds combine to form element or substances.

Combination reaction



Decomposition reaction



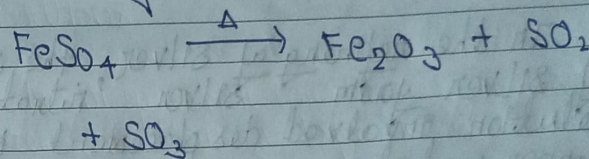
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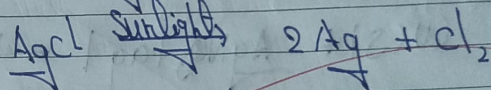
That's why it is considered as opposite of combination reaction

12. Write one equation each for decomposition reactions where energy is supplied in the form of heat, light or electricity

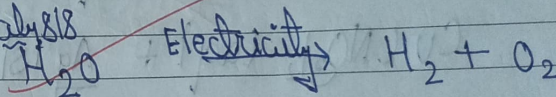
Thermolysis



Photolysis



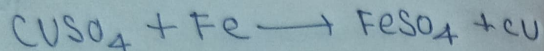
Electrolysis



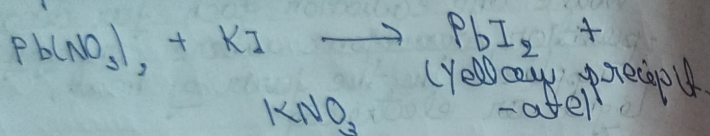
13. What is the difference between displacement and double displacement reactions?

Sol:

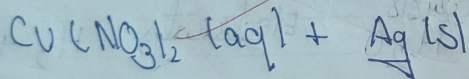
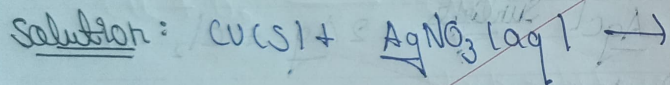
In displacement reaction the metal displaces other element on the basis of reactivity series



In double-displacement two compounds reacts to form 2 compounds by mutual exchange of ions. It is called double displacement reaction.



14, In the refining of silver, the recovery of silver from silver nitrate solution involved displacement by copper metal.
Write down the reaction involved.



Cu is more reactive than silver.

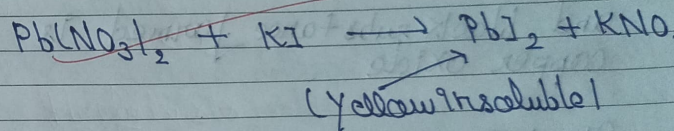
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GENIUS MIND

15. What do you mean by a precipitation reaction? Explain by giving examples.

A chemical reaction in which an insoluble substance or precipitate is formed is known as precipitation reaction.

When lead nitrate reacts with potassium iodide to form an insoluble yellow substance lead iodide and potassium nitrate.



16. Explain the following terms of gain or loss of oxygen with two examples each.

(a) Oxidation

It's the gain of oxygen & removal of hydrogen.

(b) Reduction

Gain of hydrogen &
Removal of oxygen

17. A shiny brown coloured element 'X' on heating in air becomes black in colour.

Name the element 'X' and the black coloured compound formed.

Sol: The element 'X' is copper powder

and black substance formed is copper oxide



'X'

↑
black substance
(Copper oxide)

18. Why do we apply paint on iron articles?

Sol: By applying paint on particles made up of iron, the article gets a layer of paint on the surface and surface of iron article can't react with atmospheric gases and it prevents rusting of iron.

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19. Oil and fat containing food items are flushed with nitrogen. Why?

They are flushed with nitrogen as nitrogen prevents oxidation (antioxidant). Keeping fats and oil containing food in air tight containers help to slow down oxidation.

20. Explain the following terms with one example each.

(a) Corrosion.

It is a surface deterioration / degradation of metal in which they convert into more stable form due to presence of atmospheric gases.

(b) Rancidity

The oxidation of oil or fat containing food when they get into the contact of air is known as Rancidification.

Madhusu
23/6/24